

Power Control Guide

Power Application Controller®





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1 INTRODUCTION

Power Measurement and Control is an option available in the FOC motor control firmware that allows the user to regulate motor power consumption instead of speed.

This option is specifically useful in the following cases:

- 1) Applications where power consumption from a power source needs to be limited.
- 2) Applications where speed or torque control are not desirable due to the nature of load conditions.

Some examples of applications where power control may be desirable are vacuum cleaners and garden blowers.

2 DESCRIPTION

2.1 Power Measurement Methods

The firmware project includes two methods to measure power. Users can select which method they want to use depending on their needs. The available methods are described in Table 1.

Table 1: Power Measurement Methods

Method#	Technique	Output	Advantages	Disadvantages
1	Combines the calculated instantaneous power from each individual phase	Total Instantaneous Power	Requires no extra hardware components	Not very accurate, especially in 1-shunt mode
2	Measures average bus current	Average Bus Current	Very accurate	1) Requires extra hardware components 2) Limited to topologies which use total motor current shunt

2.1.1 Method #1

2.1.1.1 Description

This method calculates the instantaneous power consumed in each phase and combines them to obtain a total instantaneous power. The total power calculated is scaled in $1/(V_{bus} * I_{phase})$ units which later needs to be descaled.

2.1.1.2 Requirements

This measurement method has no hardware requirements. No board file configuration requirements are necessary either since no new ADC resources are used.

2.1.2 Method #2

2.1.2.1 Description

This method measures total motor current and filters it through an RC filter. The resulting DC signal represents the average bus current, which is then fed to an analog IO for ADC conversion. Currently the firmware supports conversion only through AIO6.

2.1.2.2 Hardware Requirements

This method requires a shunt resistor that measures total motor current. Examples of acceptable configurations are 1-shunt or (2+Total)-shunt configurations. Please refer to Figure 1 and Figure 2 below for reference.

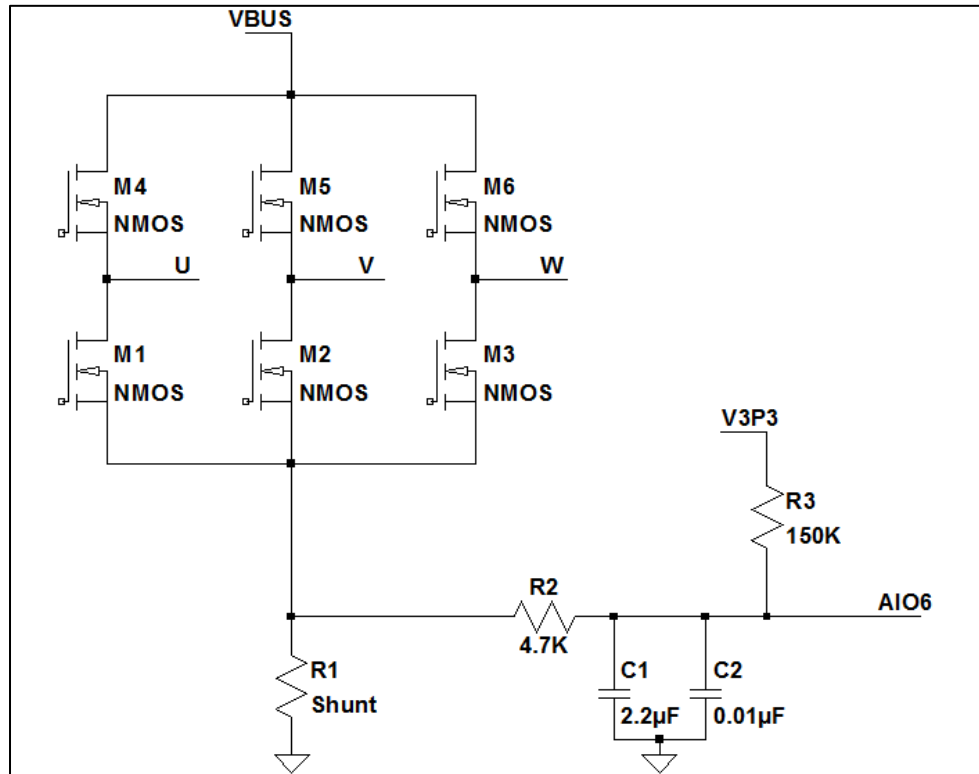


Figure 1: Average Current Measurement Circuit for 1-Shunt Configuration

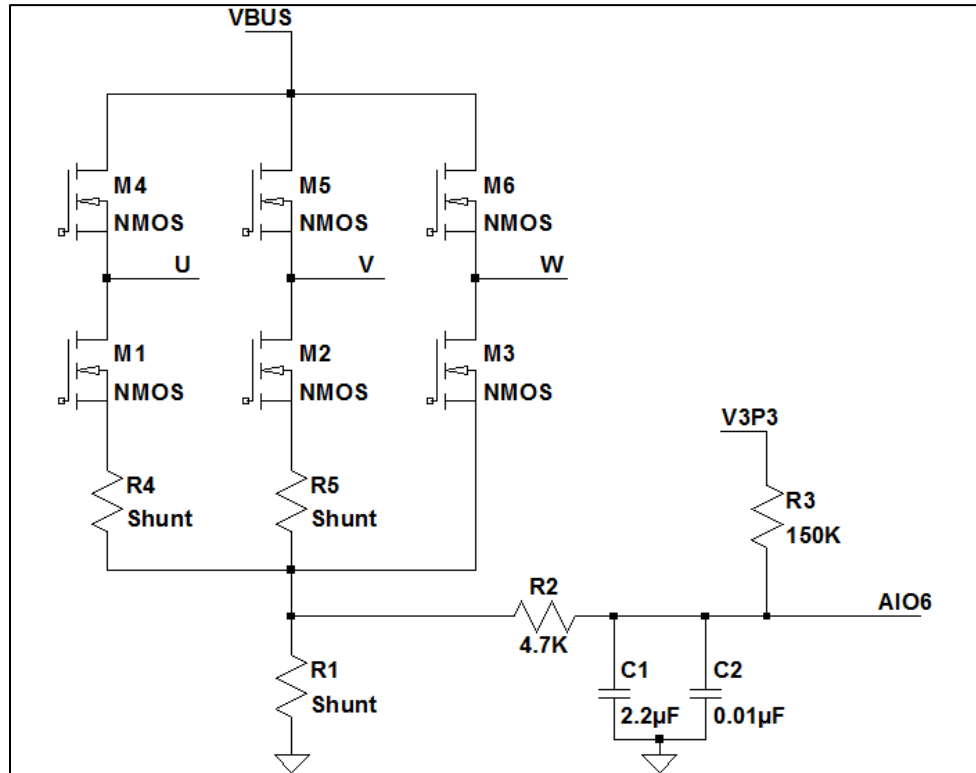


Figure 2: Average Current Measurement for 2-Shunt + Total Configuration

2.2 Firmware Flow

Power Measurement

The power calculation function first calculates the scaled power using one of the measurement methods selected by compile option. The power result is then filtered and descaled before completing. There are compile options to apply current correction or total power correction to the result before returning the final result. Please refer to Figure 3.

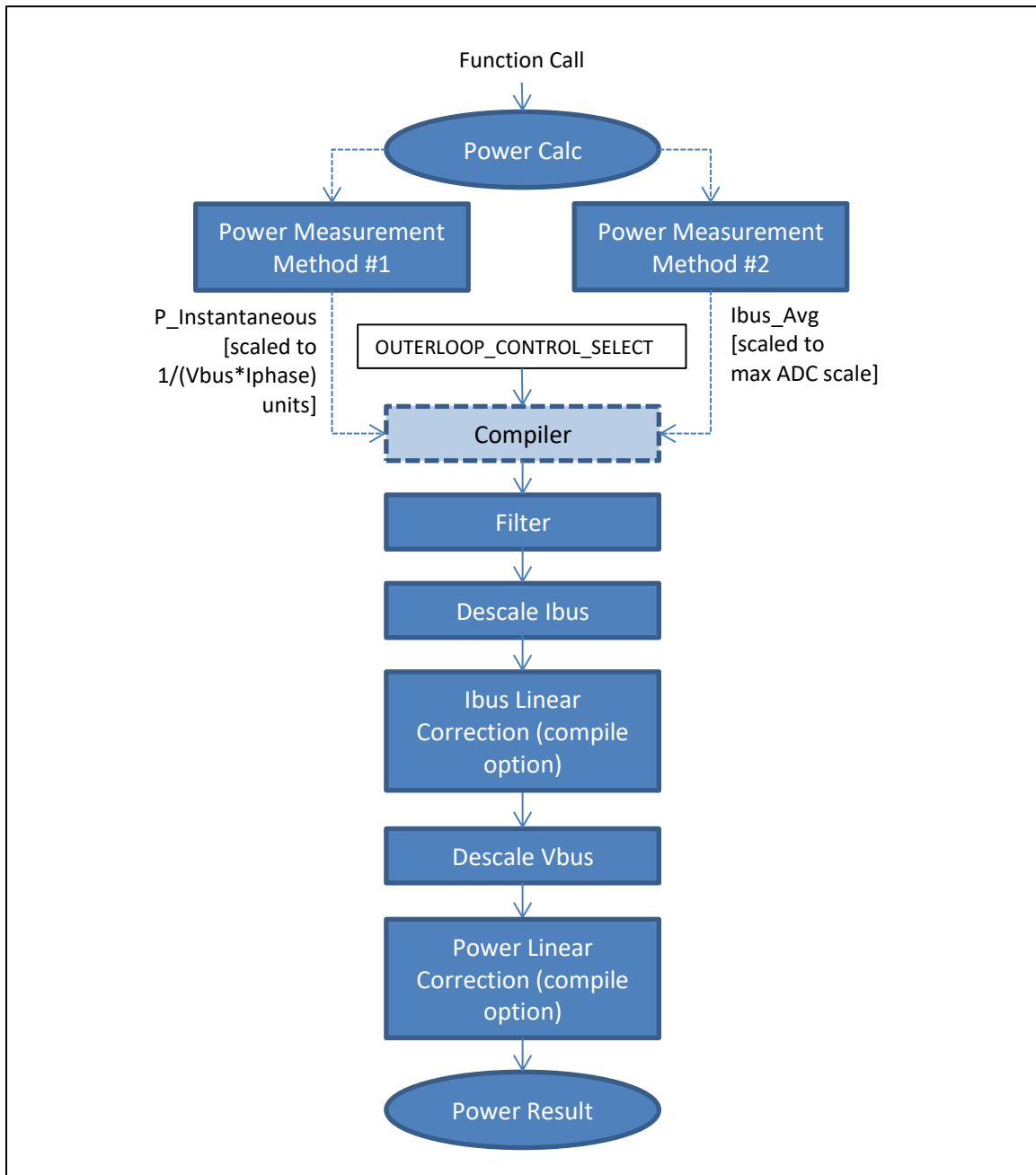


Figure 3: Power Measurement Flow Diagram

Power Control

Power control uses the result from the power calculation function and feeds it to the outer loop PI controller which then generates the Iq reference value used by the FOC control algorithm

The outer loop controller can be used as a power controller or as a speed controller depending on how the project is compiled. When the project is compiled in power control mode, speed control is no longer necessary and the outer loop PI controller functions as a power PI controller instead. Please refer to Figure 4.

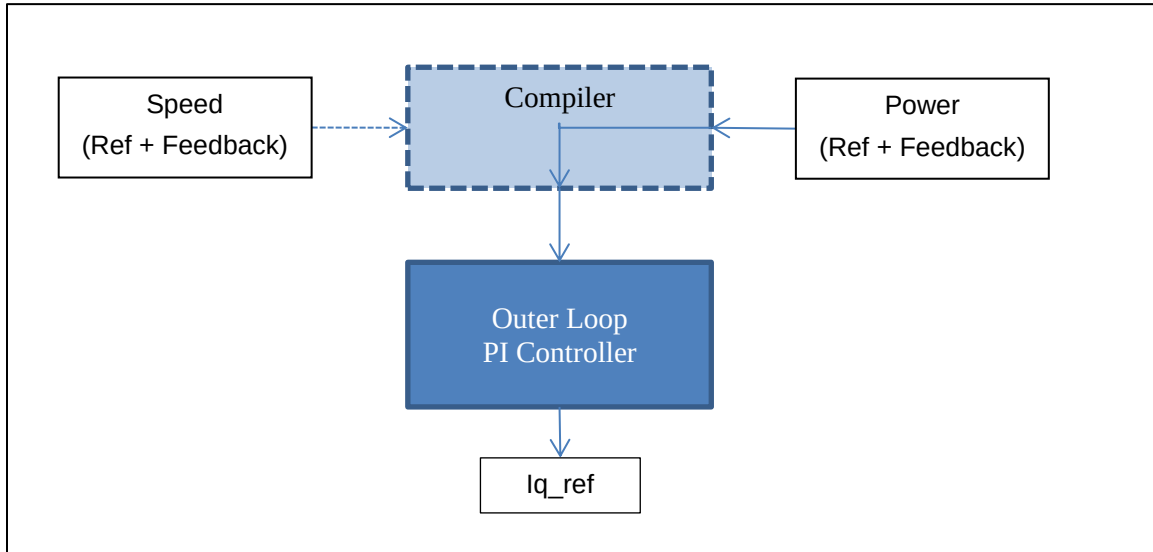


Figure 4: Outerloop PI Controller in Power Control Mode

3 USING POWER MEASUREMENT & CONTROL

3.1 Usage

To enable power measurement:

- 1) Find the `ENABLE_POWERMEASURE` option in `config_features.h` and verify that it is defined.
- 2) Find the `POWER_MEASURE_SELECT` option in `config_features.h` and verify that the desired method is selected.
 - a. If Power Control Method 1 is selected, it can be used with board files with any shunt configuration.
 - b. If Power Control Method 2 is selected, ensure that a board file is selected that supports Method 2. Example: `BOARD_EP5223LVM1_1_1SHUNT_PWR2`
- 3) Select the desired power measurement firmware options
- 4) If power control is desired, find the `OUTERLOOP_CONTROL_SELECT` option and verify that the power mode is selected.
- 5) The GUI will automatically detect that power measurement is enabled and will display the measured power in the status panel.
- 6) If power control is selected, the GUI will automatically display the Power Control panel in the Tuning view.
- 7) Enter the desired power level in the Power Setpoint box in the Operation panel.

3.2 Firmware Options

Below are the firmware options available for power measurement in `config_features.h`:

ENABLE_LCORR_IBUS

Applies a linear correction formula to the average bus current calculation as shown below:

$$I_{bus_corrected} = (M * I_{bus_calculated}) + B$$

The M and B parameters are defined respectively as `LCORR_IBUS_M` and `LCORR_IBUS_B` in `config_features.h`.

ENABLE_LCORR_PWR

Applies a linear correction formula to the power calculation as shown below:

$$Power_corrected = (M * Power_calculated) + B$$

The M and B parameters are defined respectively as `LCORR_PWR_M` and `LCORR_PWR_B` in `config_features.h`.

3.3 Firmware Parameters

Power measurement uses the firmware specified configuration parameters below. These parameters are defined in `config_features.h`:

POWER_FILTER_FREQ_CUTOFF_HZ_Q16

Defines the cutoff frequency for the single-order IIR filter used to filter the measured power.

3.4 Tuning Parameters (GUI)

The following parameters are adjustable from the GUI for SIM tuning purposes:

Kp Gain (x)

Proportional gain for PI controller regulating the power level.

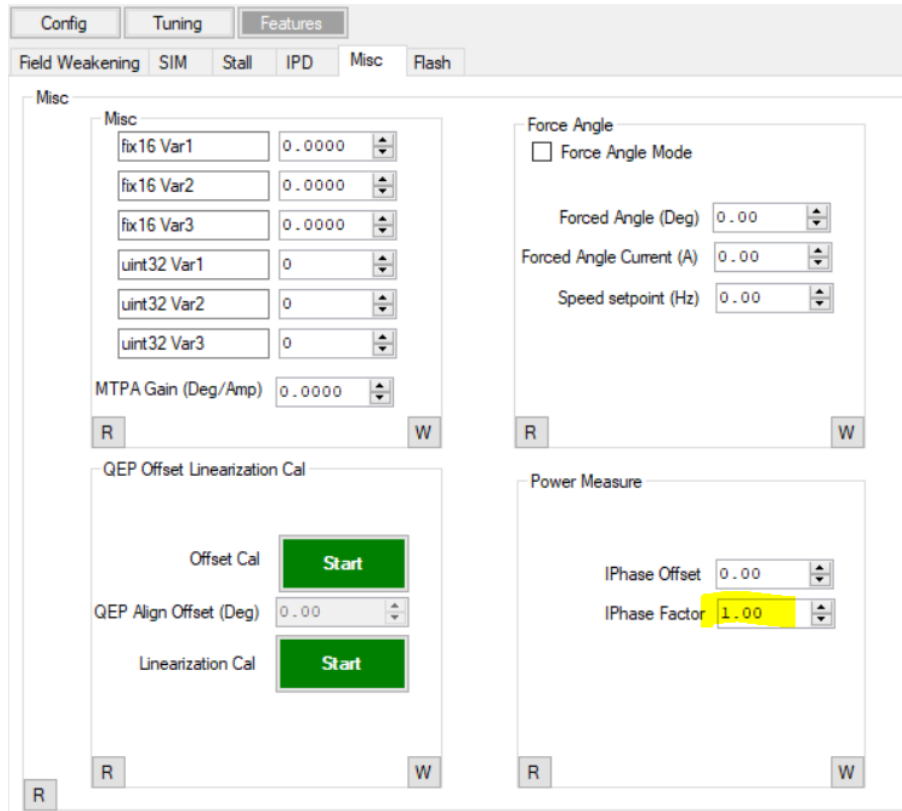
Ki Gain (x)

Integral gain for PI controller regulating the power level.

Power Ramp Rate (W/s)

Desired ramp rate for power to reach its final target.

For power measurements, please config the current measurement factor and offset as illustrated in Figure 5. The fine adjustment may be necessary for more accurate results.



The screenshot shows the 'Features' tab in the software interface, with the 'Misc' sub-tab selected. The 'Power Measure' section is highlighted, showing the 'IPhase Factor' set to 1.00. Other visible settings include 'IPhase Offset' at 0.00, 'Speed setpoint (Hz)' at 0.00, and 'Forced Angle (Deg)' at 0.00. The 'Misc' section also includes 'fix16 Var1', 'fix16 Var2', 'fix16 Var3', 'uint32 Var1', 'uint32 Var2', 'uint32 Var3', and 'MTPA Gain (Deg/Amp)'.

Figure 5: Power measurement configuration



CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

Web: www.qorvo.com

Tel: 1-844-890-8163

Email: customer.support@qorvo.com

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